

Practical No.1

Aim: To Study of different tools available for Data Mining.

Theory: Rapidminer Studio is a comprehensive data science platform with visual workflow design and full automation. It means that we don't have to do the coding for data mining tasks. Rapidminer Studio is one of the most popular data science tools.

This is the graphical user interface of the blank process in RapidMiner Studio. It has the repository that holds our dataset. We can import our own datasets. It also offers many public datasets that we can try.

RapidMiner Studio uses a client/server model with the server offered either on-premises or in public or private cloud infrastructures.

RapidMiner Studio provides data mining and machine learning procedures including: data loading and transformation (ETL), data preprocessing and visualization, predictive analytics and statistical modeling, evaluation, and deployment. RapidMiner Studio is written in the Java programming language. RapidMiner Studio provides a GUI to design and execute analytical workflows. Those workflows are called "Processes" in RapidMiner Studio and they consist of multiple "Operators". Each operator performs a single task within the process, and the output of each operator forms the input of the next one.

WEKA - an open source software provides tools for data preprocessing, implementation of several Machine Learning algorithms, and visualization tools so that you can develop machine learning techniques and apply them to real-world data mining problems. Weka contains a collection of visualization tools and algorithms for data analysis and predictive modelling, together with graphical user interfaces for easy access to these functions. The original non-Java version of Weka was a Tcl/Tk front-end to (mostly third-party) modelling algorithms implemented in other programming languages, plus data preprocessing utilities in C and a makefile-based system for running machine learning experiments.

Weka supports several standard data mining tasks, specifically, data preprocessing, clustering, classification, regression, visualization, and feature selection. Input to Weka is expected to be formatted according to the Attribute-Relational File Format and filename with the .arff extension. All Weka's techniques are predicated on the assumption that the data is available as one flat file or relation, where a fixed number of attributes describes each data point (numeric or nominal attributes, but also supports some other attribute types). Weka provides access to SQL databases using Java Database Connectivity and can process the result returned by a database query. Weka provides access to deep learning with Deeplearning4j.

Conclusion: In this way, we have studied about different tools for Mining Data.